

GA4 E-commerce Analytics: Revenue, Funnel & Growth Diagnostics

💡 KEY BUSINESS TAKEAWAYS

- Organic Search & Referral channels drive 50.6% of total monthly revenue (\$73.03K / \$144.31K).
- Non-Organic Paid Campaigns show high exit rates (~49.8%), signaling landing page or targeting mismatches.
- 75% of landing traffic drops before viewing a product; fixing navigation will unlock high-intent conversion volume.

Executive Overview & Strategic Action Plan

• Business Health & Revenue Drivers:

November performance generated \$144.31K in total revenue across 141.0K sessions and 1,618 purchases. Revenue is heavily concentrated in Organic Search (\$40.29K) and Referral channels (\$32.74K), maintaining healthy average order values (\$87.83 to \$90.83).

• Core Business Bottlenecks:

- Landing Page Drop-off: Over 75% of users exit between initial site entry (session_start: 106.5K) and product interaction (view_item: 26.0K), indicating friction on category and homepage navigation.
- Paid Traffic Mismatch: Non-organic paid campaigns exhibit high bounce rates (up to 49.79%), burning ad spend on low-intent entrances.

• Strategic Action Plan:

1. UX/Path Optimization: Streamline homepage hero banners and navigation links to drive cold landing traffic directly into product detail pages (view_item).
2. Budget Reallocation: Shift marketing spend from underperforming paid campaigns into top-converting referral channels and high-intent SEO landing pages.

📌 QUICK DASHBOARD NAVIGATION

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campaign ▾

Select date ▾

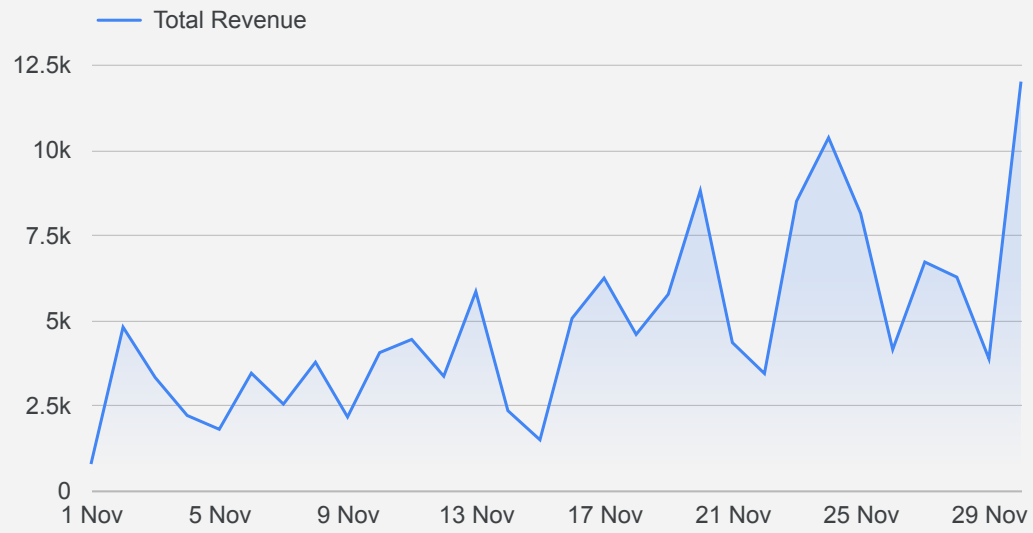
[View Raw SQL on GitHub](#)

Business & Campaign Performance

Total Revenue
\$144.26k

Average Order Value
\$87.83

total_sessions
108.7k



	campaign	Total Sessions	Unique Purchase Sessions	Total Revenue ▾	Average Order Value
1.	(organic)	36,136	428	40,293	88.27
2.	(referral)	19,854	365	32,743	90.83
3.	(direct)	25,127	351	31,133	88.16
4.	(data deleted)	7,946	240	21,634	92.69
5.	<Other>	19,625	234	18,457	79.2

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	campaign	avg_items_per_order ▾
1.	(organic)	183.1
2.	(data deleted)	171.5
3.	(referral)	166.3
4.	(direct)	145.7
5.	<Other>	143.9

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Executive Insight

Performance Snapshot: Revenue reached \$144.26K across November 2020. Organic Search and Referral were top drivers, generating \$40.29K and \$32.74K respectively with average order values above \$88.00.

Strategic Action: Organic traffic exhibits high purchasing intent. Allocate marketing focus toward optimizing SEO landing pages and top referral channels to sustain low-CAC revenue growth.

Marketing Acquisition & Session Quality

Total Entrances
71.8k

Total Page Views
453.9k

Bounce Rate
43.51%

campaign	Total sessions	Unique Purchase Sessions	Bounce Rate
1.. (organic)	72,272	428	41.42%
2.. (direct)	50,254	351	42.37%
3.. (referral)	39,708	365	43.78%
4.. <Other>	39,250	234	40.19%
5.. (data deleted)	15,892	240	49.79%

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	Page Path	Total Users	Entrances	Page Views
1.	https://shop.go...	40,065	22,215	57,668
2.	https://google...	14,174	10,725	18,242
3.	https://www.go...	9,199	7,425	11,169
4.	https://shop.go...	9,072	6,686	10,622
5.	https://shop.go...	9,462	3,688	12,523
6.	https://shop.go...	2,490	2,032	3,015
7.	https://shop.go...	10,928	1,865	21,685

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Acquisition & Landing Page Insights

Channel Health: Organic Search is your primary volume driver (72,272 sessions, 428 purchases) with the lowest bounce rate (41.42%). Referral traffic drives the highest purchase efficiency relative to its volume (365 purchases across 39.7K sessions).

Landing Page Engagement: Primary entry URLs (/home, top brand pages) capture over 60% of total site entrances (22.2K entrances on the main store page).

Strategic Action: Reallocate budget toward high-converting referral pathways and optimize high-traffic organic landing pages with clearer checkout CTAs to capture engaged intent.

E-commerce Conversion Journey & Technical Diagnostics

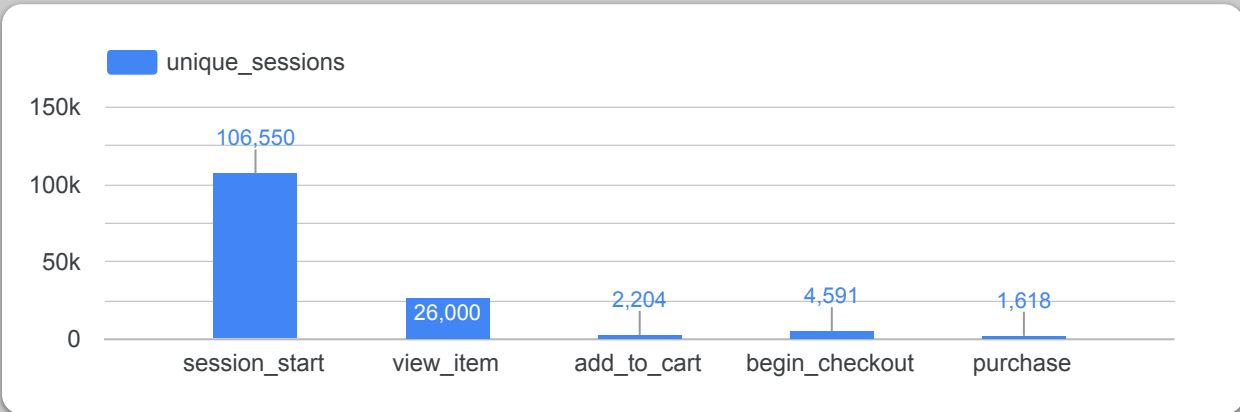
Total Funnel Sessions
141.0k

Completed Purchases
1,618

Overall Funnel Conversion Rate
0.06%

Item Revenue
\$144.31k

	Item Name	Items Added to cart	Item Purchased	Item Revenue ▾
1.	Google Zip Ho...	147	130	6,660
2.	Google Badge...	60	126	6,091
3.	Google Men's ...	17	62	4,654
4.	Google Crewn...	58	96	4,312
5.	Super G Unise...	193	133	4,102
6.	Google F/C Lo...	0	143	3,630
7.	Google Crewn...	50	72	3,245
8.	Google Navy ...	118	111	2,754
9.	Google Wome...	16	29	2,714
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	Page Path (404 or Error)	Page Views ▾	Total Users
1.	https://shop.googlemerchandisest...	2	2
2.	https://shop.googlemerchandisest...	1	1
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Conversion & Technical Diagnostic Insights

Funnel Performance: The store recorded 141.0K total funnel sessions and 1,618 completed purchases, driving \$144.31K in total item revenue with an overall session conversion rate of 0.06%.

Drop-off Bottleneck: The largest conversion leak occurs between initial landing (106.5K session_start hits) and product engagement (26.0K view_item hits), where over 75% of users exit before viewing a specific item.

Item Performance & Error Diagnostics: Product sales are led by apparel items like the Google Zip Hoodie (\$6,660 revenue, 130 purchases). Technical tracking successfully isolated low-volume 404/error page hits, ensuring broken paths are identified before impacting high-volume conversion flows.

Strategic Action: Optimize homepage navigation and primary category banners to seamlessly push landing page traffic into product detail pages (view_item), directly addressing the primary 75% traffic drop-off point.

Data Architecture, SQL Pipeline & Technical Appendix

Appendix: Data Pipeline & Technical Architecture

1. Data Source & Pipeline

- **Source Dataset:** Public Google Analytics 4 (**GA4**) E-commerce dataset hosted in **Google BigQuery**.
- **Reporting Period:** November 1, 2020 – November 30, 2020.
- **Pipeline Architecture:** GA4 Raw Event Export → BigQuery SQL Transformation & Aggregation → Looker Studio Visualizations.

2. SQL Transformation Strategy

- Nested GA4 event parameters (event_params, user_properties) were unnested and structured into **3 specialized Master Views** using SQL UNION ALL constructs.
- **Why this matters:** Prevents metric duplication and row multiplication across page-level vs. session-level dimensions, maintaining absolute data accuracy across all pages.

3. Master Views Reference

- v_business_performance: Tracks campaign ROI, revenue metrics, session totals, and bounce rates.
- v_ux_and_page_analytics: Captures landing page entrances, total page views, user counts, and 404/error page paths.
- v_funnel_analytics: Computes sequential e-commerce funnel counts (session_start → purchase) and item-level revenue.

4. Code Repository

- **Full Transformation Queries:** Access all raw SQL scripts and view definitions on GitHub:

 [View Complete SQL Repository on GitHub](#)